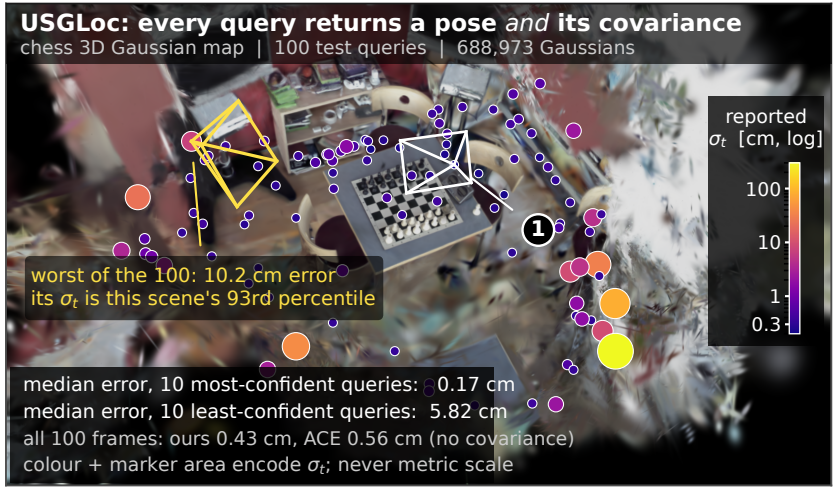
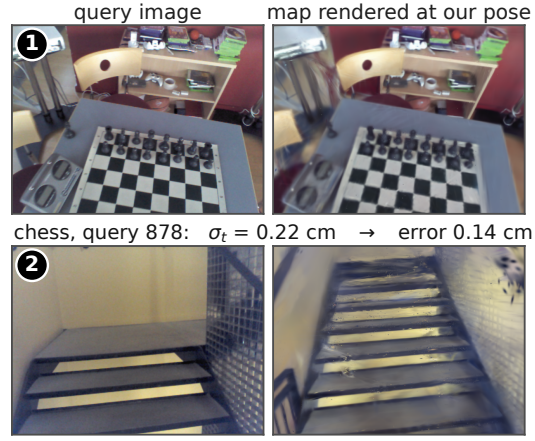

USGLoc: every query returns a pose and its covariance
chess 3D Gaussian map | 100 test queries | 688,973 Gaussians
reported
σ_t [cm, log]
100
10
1
0.3
1
worst of the 100: 10.2 cm error
its σ_t is this scene's 93rd percentile
median error, 10 most-confident queries: 0.17 cm
median error, 10 least-confident queries: 5.82 cm
all 100 frames: ours 0.43 cm, ACE 0.56 cm (no covariance)
colour + marker area encode σ_t; never metric scale


query image
map rendered at our pose
1
chess, query 878: σ_t = 0.22 cm → error 0.14 cm
2
stairs, query 494: σ_t = 1160 cm → error 144 cm
Pose 2 is unusable − and USGLoc says so first.
No prior 3DGS localizer reports a pose covariance.